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Electronic Fundamentals Exam

Exercise 1 (6pts)

Consider the circuit presented in **Figure 1**. A load resistor R_4 is connected between the output terminals **A** and **B**.

Given: $E_1 = 4 \text{ V}$, $R_1 = 16 \Omega$, $R_2 = 4 \Omega$, $R_3 = 6 \Omega$, and $R_4 = 24 \Omega$.

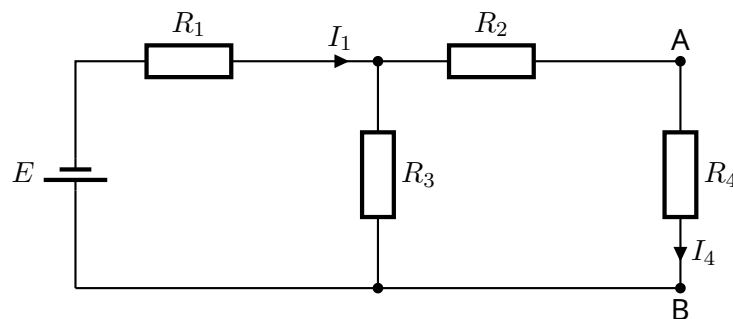


Figure 1

1. Determine the expression and calculate the equivalent resistance R_{eq} as a function of R_1, R_2, R_3 and R_4 .

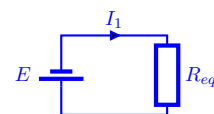
$$R_2 \text{ en serie } R_4 \Rightarrow R_{24} = R_2 + R_4 = 28 \Omega$$

$$R_{24} \parallel R_3 \Rightarrow R' = \frac{R_{24}R_3}{R_{24}+R_3} = \frac{28 \cdot 6}{34} = 4.94 \Omega$$

$$R_{eq} = R_1 + R' = 4.94 + 16 \Rightarrow \boxed{R_{eq} = 20.94 \Omega}$$

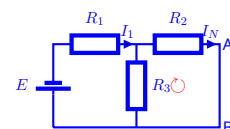
2. Calculate the current I_1 using Kirchhoff's Voltage Law (loop law).

$$I_1 = \frac{E}{R_{eq}} = \frac{4}{20.94} \Rightarrow \boxed{I_1 = 190 \text{ mA}}$$



3. Using Figure 1 and applying Norton's theorem, determine the expression of the Norton current I_N in terms of I_1 , R_2 , and R_3 as seen across the load resistor R_4 .

$$\text{Current divider: } I_N = \frac{R_3}{R_3+R_2} \cdot I_1$$

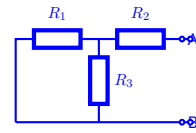


4. Calculate the value of the Norton current I_N .

$$I_N = \frac{6}{10} \cdot 190 \Rightarrow \boxed{I_N = 114 \text{ mA}}$$

5. Give the expression of Norton resistance R_N and calculate its numerical value.

$$R_N = \frac{R_1 \cdot R_3}{R_1 + R_3} + R_2 \Rightarrow \boxed{R_N = 8.36 \, \Omega}$$



$$V_{CC} = \left(I_C + \frac{I_C}{\beta} \right) (R_E + R_C) + V_{CE}$$

$$I_C = \frac{V_{CC} - V_{CE}}{(R_E + R_C) \left(1 + \frac{1}{\beta} \right)}$$

6. Determine the saturation current $I_{C_{sat}}$ corresponding to the condition $V_{CE} = 0V$.

$$V_{CE} = 0 \Rightarrow I_{sat} = \frac{V_{CC}}{(R_E + R_C) \left(1 + \frac{1}{\beta} \right)} \Rightarrow I_{sat} \approx 9.09 \text{ mA}$$

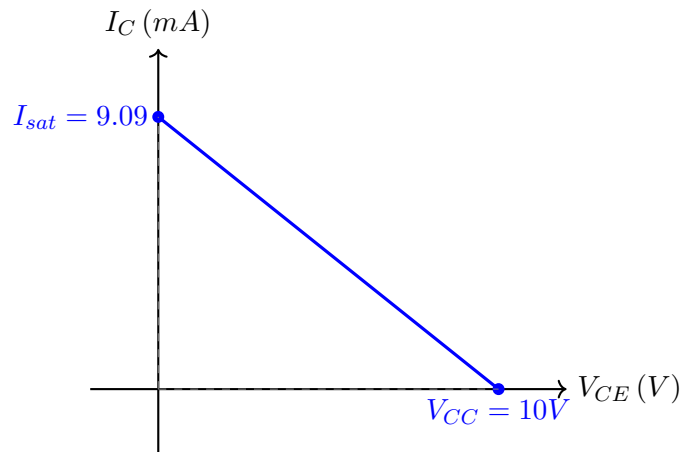
7. Determine the cutoff voltage V_{CE} corresponding to the condition $I_C = 0A$.

$$I_C = 0 \Rightarrow V_{CE} = V_{CC} \Rightarrow V_{CE} = 10 \text{ V}$$

8. Establish the equation of the input line $I_B = f(V_{BE})$ (input characteristic).

$$\begin{aligned} V_{CC} &= (R_C + R_E)I_E + R_B I_B + V_{BE} \\ &= (I_B + \beta I_B)(R_C + R_E) + R_B I_B + V_{BE} \\ \Rightarrow I_B &= \frac{V_{CC} - V_{BE}}{R_B + R_C + R_E(1 + \beta)} \end{aligned}$$

9. Plot the DC load line $I_C = f(V_{CE})$ (output characteristic).



10. Calculate the values of I_C and I_B required so that $V_{CE} = 5 \text{ V}$, while keeping the other parameters unchanged.

$$\textcircled{0,5} \quad I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E} = \frac{10 - 5}{1.1} \Rightarrow I_C = 4.54 \text{ mA}$$

$$\textcircled{0,25} \quad I_B = \frac{I_C}{\beta} \Rightarrow I_B = 69.84 \mu\text{A}$$

Given:

$\beta = 65$, $V_{CC} = 10\text{V}$, $V_{BE} = 0.7\text{V}$, $R_B = 17 \text{ k}\Omega$, $R_C = 1 \text{ k}\Omega$, and $R_E = 100 \Omega$.